

**Sri Sivasubramaniya Nadar College of Engineering, Chennai**  
(An autonomous Institution affiliated to Anna University)

|                     |                                                    |          |           |
|---------------------|----------------------------------------------------|----------|-----------|
| Degree & Branch     | M.Tech (Integrated) Computer Science & Engineering | Semester | V         |
| Subject Code & Name | ICS1512 – Machine Learning Algorithms Laboratory   |          |           |
| Academic Year       | 2025–2026 (Odd)                                    | Batch    | 2024–2029 |

**Experiment 8: Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Clustering**

### Objective

To implement and analyze the performance of clustering algorithms on the Human Activity Recognition dataset:

- **Model A:** K-Means Clustering.
- **Model B:** DBSCAN (Density-Based Spatial Clustering of Applications with Noise).
- **Model C:** Hierarchical Agglomerative Clustering (HAC).

Students must visualize and compare the clusters formed by each algorithm, report results, and analyze performance.

### Dataset

**Download:** [Human Activity Recognition Using Smartphones Dataset](#)

- 30 volunteers (aged 19–48). - 6 activities: WALKING, WALKING\_UPSTAIRS, WALKING\_DOWNSTAIRS, SITTING, STANDING, LAYING. - Sensors: 3-axis accelerometer and gyroscope (50 Hz). - Preprocessed into windows of 2.56s (128 readings/window) with feature extraction from time and frequency domain.

### Theory

#### K-Means Clustering

- Objective: Partition data into  $k$  clusters by minimizing within-cluster variance. - Distance metric: Euclidean. - Algorithm:

1. Initialize  $k$  centroids.
2. Assign points to nearest centroid.
3. Update centroids as mean of assigned points.
4. Repeat until convergence.

Table 1: K-Means Elbow Method Results

| Number of Clusters ( $k$ ) | WCSS (Inertia) | Silhouette Score |
|----------------------------|----------------|------------------|
| 2                          | ----           | ----             |
| 3                          | ----           | ----             |
| 4                          | ----           | ----             |
| 5                          | ----           | ----             |
| 6                          | ----           | ----             |
| 7                          | ----           | ----             |
| 8                          | ----           | ----             |

### Elbow Method for Choosing $k$

- Compute **Within-Cluster Sum of Squares (WCSS)** for different  $k$  values. - Plot  $k$  vs. WCSS.
- The “elbow point” (where reduction in WCSS slows down) suggests the optimal  $k$ .

#### Students must:

- Fill in the table above after running experiments.
- Plot  $k$  vs. WCSS (Elbow curve).
- Plot  $k$  vs. Silhouette Score.
- Select and justify the best  $k$ .

### DBSCAN

- Density-based clustering algorithm. - Parameters:  $\epsilon$  (neighborhood radius),  $minPts$  (minimum points). - Core points, border points, and noise are identified. - Advantage: Can discover clusters of arbitrary shape and handle noise.

### Hierarchical Clustering

- Builds hierarchy of clusters using either:
  - Agglomerative (bottom-up merging).
  - Divisive (top-down splitting).
- Linkage criteria: Single, Complete, Average, Ward’s method. - Represented by a dendrogram.

### Steps for Implementation

1. Load the dataset and preprocess:
  - Handle missing values.
  - Encode categorical labels.
  - Apply normalization/standardization.

2. Perform exploratory data analysis (EDA):
  - Visualize feature distributions.
  - Apply dimensionality reduction (PCA/t-SNE) for visualization.
3. Apply clustering algorithms:
  - **K-Means:** Use Elbow method to choose  $k$ .
  - **DBSCAN:** Tune  $\epsilon$  and  $minPts$ .
  - **Hierarchical:** Apply Agglomerative clustering with Ward's linkage.
4. Visualize clusters in 2D using PCA/t-SNE.
5. Compare results with ground truth activity labels (although clustering is unsupervised).
6. Analyze performance using clustering metrics.

## Evaluation Metrics

- **Internal:** Silhouette Score, Davies–Bouldin Index, Calinski–Harabasz Index.
- **External (w.r.t. true labels):** Adjusted Rand Index (ARI), Normalized Mutual Information (NMI).
- Confusion matrix (optional, by mapping clusters to activities).

## Graphs and Visualization

Students should produce:

- Elbow curve:  $k$  vs. WCSS.
- Silhouette curve:  $k$  vs. Silhouette Score.
- Scatter plots of clusters (after PCA/t-SNE).
- Dendrograms for hierarchical clustering.
- Bar plots of evaluation metrics across algorithms.

## Observation Questions

- Which algorithm produced the most meaningful clusters? Why?
- How sensitive was K-Means to the choice of  $k$ ?
- Did DBSCAN detect noise or small clusters effectively?
- How does linkage choice (single/complete/ward) affect hierarchical clustering?
- Which internal metric best matched your visual intuition of cluster quality?

## Report Checklist

- Aim and Objective
- Preprocessing Steps
- K-Means with Elbow Method Results
- Clustering Implementation (K-Means, DBSCAN, Hierarchical)
- Evaluation Metrics and Results
- Graphs and Visualizations
- Observations and Comparative Analysis